



URBAN PLANNING PORTFOLIO

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Grounded in transport, driven by change

I graduated with a Masters in City & Regional Planning at Cornell University this August, with a certification in Infrastructure Policy, Management & Finance. Driven by a strong interest in transportation systems, infrastructure, and economic development, I am particularly interested in how cities can grow in ways that are both economically resilient and socially sustainable, while responding to future urban challenges.

I'm especially drawn to how rapid urbanization reshapes cities, the congestion it creates, the innovative mobility solutions it demands, and the need for transport and energy solutions tailored to each region's specific growth patterns.

My academic foundation includes a double major in Economics & Political Science, where I built rigorous analytical training and a grounding in how capital allocation and public policy shape development outcomes.

TOOLKIT

- ArcGIS / QGIS
- Python
- Stata · R
- GTFS Analysis
- Infrastructure Finance Modeling

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Projects

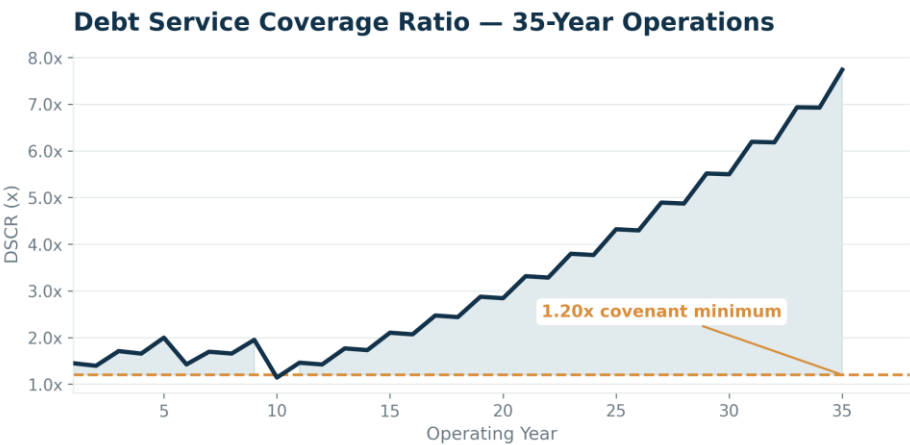
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Transit Infrastructure Finance Interborough Express, NYC

Capital Stack — \$5.54B Total



Sources of funds, Interborough Express (IBX) Light Rail



PROJECT SUMMARY

Developing a financing and implementation framework for a \$5.5B outer-borough light rail corridor — evaluating capital structure across state, federal, bond, and value-capture funding scenarios.

\$529M

RRIF loan: 4.5% fixed, 30-yr amortization, 5-yr I/O deferral

1.20x

DSCR covenant minimum (1.35x target)

\$2.56B

Net cash flow, 35-yr operating period

Ridership sensitivity: a 20% downside turns Year 1 NOI negative — underscoring the need for an MTA operating subsidy backstop — while a 20% upside lifts DSCR to 4.33x, climbing past 7x by Year 35.

Faster Roads, Fewer Refuges

Motorcycle Exposure on Lahore's Signal-Free Corridors · Lahore, Pakistan · 2026

+50%

node increase
post-SFC

10.67

vs 8.63 mean
exposure score

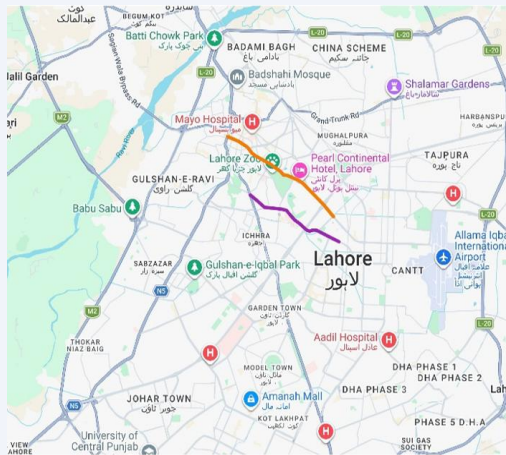
65%

higher mean speed
Jail vs Mall Rd

Jail Road — score by node



Node Exposure Score by node · Jail Road vs Mall Road mean



Study corridors in urban context



Annotated satellite node audit · Jail Road

WHAT

A comparative node-based spatial analysis of how Signal-Free Corridor design restructures motorcycle exposure on Lahore's arterial roads.

HOW

Node Exposure Score framework · Google Earth Pro audit · Distance Matrix API speed collection · PCU-controlled corridor comparison

FINDING

SFC design concentrates conflict at high-speed nodes — compounding motorcycle exposure rather than distributing it evenly.

Solar-Powered Microtransit Hubs Los Angeles

A replicable framework for siting solar-powered microtransit hubs in Los Angeles — integrating spatial suitability, solar performance modeling, and deployment feasibility. Recommended approach: phased pilots in low-deficit zones, equity-weighted siting, zoning reform, and distributed storage.

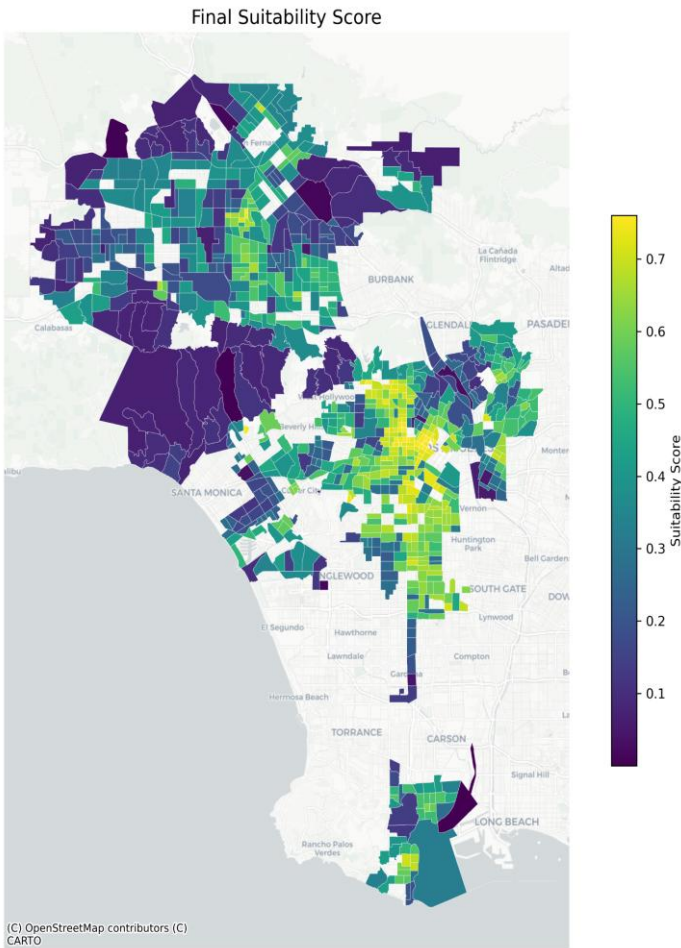


Fig. 1 — Composite suitability score by census tract, integrating zoning, population density, vehicle ownership, and transit proximity.

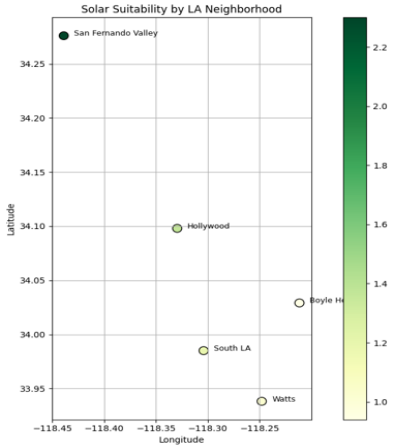


Fig. 2 — Solar suitability by neighborhood.

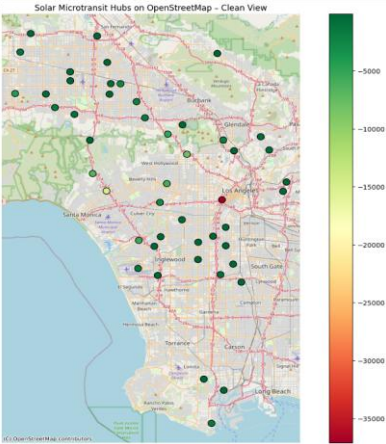


Fig. 3 — Net daily solar surplus/deficit by hub.

24 kWh/day

Was the lowest energy deficit in the region shown by *San Fernando Valley & South LA*

2,680 kWh/day

Solar deficit displayed by Hollywood, even though it has the highest transit demand

2 of 5

Pilot neighborhoods energy-balanced under baseline solar sizing (*San Fernando Valley & South LA*)

Household Energy Transitions

Clean Cooking Fuel Adoption, Cameroon

PROJECT SUMMARY

Analyzed determinants of clean cooking fuel adoption in Cameroon using nationally representative DHS data and binary logistic regression in R, examining how infrastructure, wealth, and geography shape household energy transitions.

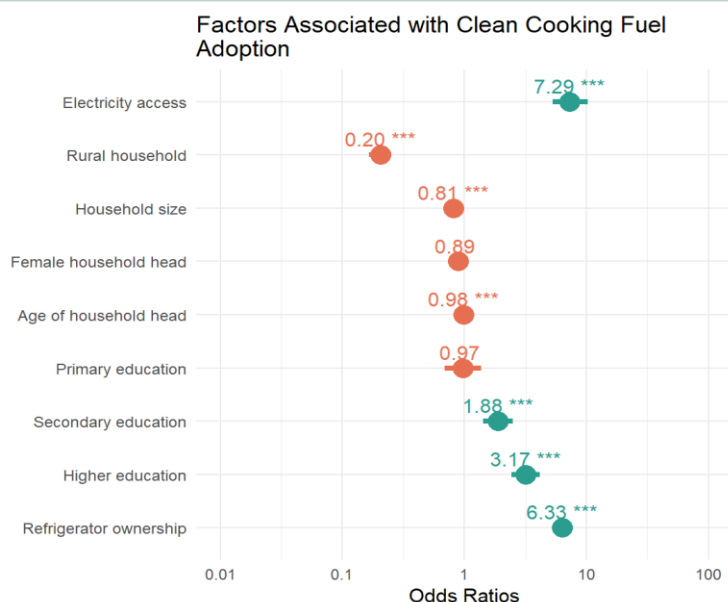


Fig. 1 — Odds ratios from logistic regression on clean cooking fuel adoption. Electricity access and refrigerator ownership are the strongest predictors — electrified households are over 7x more likely to adopt clean fuels.

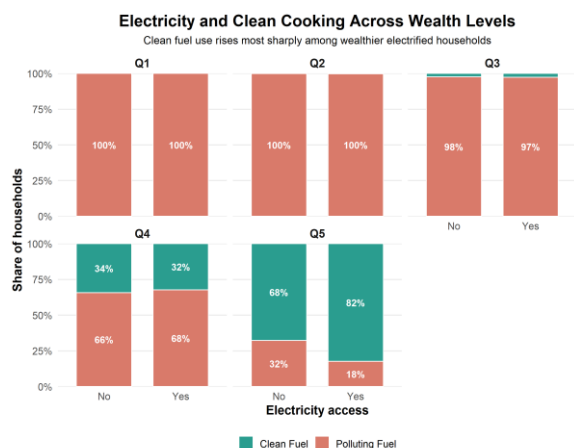


Fig. 2 — Clean fuel use by wealth quintile and electricity access. Adoption rises sharply only once households are both wealthier and electrified.

DHS Survey Clusters by Urban–Rural Status
Urban clusters are concentrated around major settlement areas in Cameroon

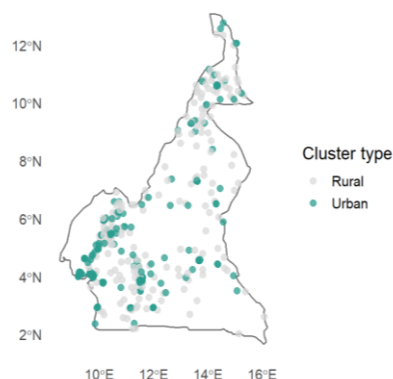


Fig. 3 — DHS survey clusters by urban–rural status. Urban clusters concentrate near major settlement areas, reflecting the electrification gap driving rural fuel choice.

Urban Design Communication

Poster Composition

The Piazza Verdecità: A Sustainable Heritage Haven

Mehr-un-risa Amin (ms2428)



Piazza Verdecità is a 22-acre mixed-use development that serves as a vibrant public gateway between a rapidly evolving urban core and a well-preserved historical district. Inspired by the grand piazzas of Europe, the innovative community seamlessly integrates residential, commercial, and civic functions within a lush, pedestrian-focused plaza. The site's proximity to the adjacent historic neighborhood was a key driver in the architectural aesthetic, which embraces a contemporary interpretation of classical Mediterranean and Neo-Classical styles. Arched entryways, columned facades, and ornamental detailing evoke a sense of timeless elegance, allowing the new development to harmonize with the existing historic buildings and streetscapes.



Figure 5: Conceptual Rendering of the Piazza Verdecità Public Plaza from its west entrance



Figure 6: Conceptual Rendering of the fountain node at the Piazza Verdecità Public Plaza



Figure 7: Aerial view of the Piazza Verdecità showing its holistic approach to blending architecture, landscape, and public amenities to create a vibrant, sustainable, and timeless urban gathering space.

Final studio poster for Graphics Communication: planned and rendered *The Piazza Verdecità*, a mixed-use development anchored by a pedestrian plaza. Produced a full site plan, land use/circulation/framework diagrams, and 3D rendered perspectives, then presented the work to an external jury.

LET'S CONNECT

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